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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xtmrctr.h"
#include "xparameters.h"

XTmrCtr tmr;

void tmr_init(){
    int status = XTmrCtr_Initialize(&tmr,XPAR_AXI_TIMER_0_DEVICE_ID );

    if(status == XST_SUCCESS)
        xil_printf("TMR INIT SUCCESSFUL\n");
    else
        xil_printf("TMR INIT FAILED\n");

    status = XTmrCtr_SelfTest(&tmr, 0);

    if(status == XST_SUCCESS)
        xil_printf("TMR SELFTEST SUCCESSFUL\n");
    else
        xil_printf("TMR SELFTEST FAILED\n");
}

int main()
{
    u32 count = 100000000*5;

    init_platform();
    tmr_init();

    XTmrCtr_Stop(&tmr, 0);
    XTmrCtr_SetResetValue(&tmr, 0, count);

    XTmrCtr_Reset(&tmr, 0);

    u32 option = XTmrCtr_GetOptions(&tmr, 0);

    //////////////////////////////////NON AUTO RELOAD MODE
    XTmrCtr_SetOptions(&tmr, 0, option | XTC_DOWN_COUNT_OPTION );

    while(1){
        XTmrCtr_Start(&tmr, 0);
        while(!XTmrCtr_IsExpired(&tmr, 0));
        XTmrCtr_Stop(&tmr, 0);
        xil_printf("Delay of 5 Sec achieved\n");
        XTmrCtr_Reset(&tmr,0);
    }

    //////////////////////////////////AUTO RELOAD MODE
    /*
    XTmrCtr_SetOptions(&tmr, 0, option | XTC_DOWN_COUNT_OPTION | XTC_AUTO_RELOAD_OPTION);

    while(1){
        XTmrCtr_Start(&tmr, 0);
        while(!XTmrCtr_IsExpired(&tmr, 0));
        XTmrCtr_Stop(&tmr, 0);
        xil_printf("Delay of 5 Sec achieved\n");
    }
    */

    cleanup_platform();
    return 0;
}

```